Lyapunov functions. From the fixed point equations (2.8) we have

$$\overline{\sigma} = \beta^{\circ}_{2n}\overline{\sigma}\left(\beta^{\circ}_{2n}\right)' + \tau^{\circ}_{2n} \geqslant \tau^{\circ}_{2n} \quad \text{and} \quad \sigma = \beta^{\circ}_{2n+1}\sigma\left(\beta^{\circ}_{2n+1}\right)' + \tau^{\circ}_{2n+1} \geqslant \tau^{\circ}_{2n+1}, \tag{A.4}$$

which, in turn, imply that

$$\overline{\sigma}_{\varpi} := \overline{\sigma}^{1/2}(I+\varpi^{-1})^{-1}\overline{\sigma}^{1/2} \leqslant \tau^{\circ}_{2n} \leqslant \overline{\sigma} \quad \text{and} \quad \sigma_{\overline{\varpi}} := \sigma^{1/2}(I+\overline{\varpi}^{-1})^{-1}\sigma^{1/2} \leqslant \tau^{\circ}_{2n+1} \leqslant \sigma. \tag{A.5}$$

Also note that if $\overline{\sigma}^{-1} \leqslant (\tau^{\circ}_{2n})^{-1}$ then $(\tau^{\circ}_{2n})^{1/2}\overline{\sigma}^{-1}(\tau^{\circ}_{2n})^{1/2} \leqslant I$. Moreover, we also readily see that

$$\begin{aligned}
&\mathbb{E}\left[(\mathcal{Z}^{\circ}_{2n}(y)-\overline{m})'\overline{\sigma}^{-1}(\mathcal{Z}^{\circ}_{2n}(y)-\overline{m})\right] \\
&= (y-\overline{m})'\left(\beta^{\circ}_{2n}\right)'\overline{\sigma}^{-1}\beta^{\circ}_{2n}(y-\overline{m}) + \mathrm{Tr}\left(\overline{\sigma}^{-1/2}\tau^{\circ}_{2n}\overline{\sigma}^{-1/2}\right) \\
&= (y-\overline{m})'\overline{\sigma}^{-1/2}\left[\left(\overline{\sigma}^{-1/2}\beta^{\circ}_{2n}\overline{\sigma}^{1/2}\right)'\left(\overline{\sigma}^{-1/2}\beta^{\circ}_{2n}\overline{\sigma}^{1/2}\right)\right]\overline{\sigma}^{-1/2}(y-\overline{m}) + \mathrm{Tr}\left(\overline{\sigma}^{-1/2}\tau^{\circ}_{2n}\overline{\sigma}^{-1/2}\right),
\end{aligned}$$

which implies

$$\begin{aligned}
\mathbb{E}\left[(\mathcal{Z}^{\circ}_{2n}(y)-\overline{m})'\overline{\sigma}^{-1}(\mathcal{Z}^{\circ}_{2n}(y)-\overline{m})\right] &= (y-\overline{m})'\overline{\sigma}^{-1/2}\left[I-\upsilon_{2n}\right]^2\overline{\sigma}^{-1/2}(y-\overline{m}) \\
&\quad + \mathrm{Tr}\left(\overline{\sigma}^{-1/2}\tau^{\circ}_{2n}\overline{\sigma}^{-1/2}\right) \\
&\leqslant (1+\lambda_{\mathsf{min}}(\varpi))^{-2}\,(y-\overline{m})'\overline{\sigma}^{-1}(y-\overline{m}) + d.
\end{aligned} \tag{A.6}$$

We are now in position to prove Proposition 5.2.

Proof of Proposition 5.2. Consider the function

$$\psi(y) := 1 + (y-\overline{m})'\overline{\sigma}^{-1}(y-\overline{m}).$$

The estimate in (A.6) ensures that

$$\mathcal{S}^{\circ}_{2n}(\psi)(y) = \mathbb{E}\left[\psi(\mathcal{Z}^{\circ}_{2n}(y))\right] \leqslant (1+\lambda_{\mathsf{min}}(\varpi))^{-2}\,\psi(y) + \left(\left(1-(1+\lambda_{\mathsf{min}}(\varpi))^{-2}\right)+d\right).$$

Next, let us observe that

$$\begin{aligned}
\mathcal{S}^{\circ}_{2n}(y_1,dy_2) &= \frac{1}{\sqrt{\det(2\pi\tau^{\circ}_{2n})}} \\
&\times \exp\left(-\frac{1}{2}\,\left((y_2-\overline{m})-\beta^{\circ}_{2n}(y_1-\overline{m})\right)\right)'\left(\tau^{\circ}_{2n}\right)^{-1}\left((y_2-\overline{m})-\beta^{\circ}_{2n}(y_1-\overline{m})\right)\right)dy_2
\end{aligned}$$

and

$$\begin{aligned}
&\left((y_2-\overline{m})-\beta^{\circ}_{2n}(y_1-\overline{m})\right)\right)'\left(\tau^{\circ}_{2n}\right)^{-1}\left((y_2-\overline{m})-\beta^{\circ}_{2n}(y_1-\overline{m})\right)\right) \\
&\leqslant 2\,(y_2-\overline{m})'\left(\tau^{\circ}_{2n}\right)^{-1}(y_2-\overline{m}) + 2\,\left(\beta^{\circ}_{2n}(y_1-\overline{m})\right)\right)'\left(\tau^{\circ}_{2n}\right)^{-1}\left(\beta^{\circ}_{2n}(y_1-\overline{m})\right)\right) \\
&\leqslant 2\,(y_2-\overline{m})'\,\overline{\sigma}^{-1}_{\varpi}\left((y_2-\overline{m})\right) + 2\,\left(1+\lambda_{\mathsf{min}}(\varpi^{-1})\right)\,(y_1-\overline{m})'\left(\beta^{\circ}_{2n}\right)'\overline{\sigma}^{-1}\beta^{\circ}_{2n}\,(y_1-\overline{m}),
\end{aligned}$$

where the second inequality follows from (A.5). Moreover, arguing again as above we arrive at

$$\begin{aligned}
&\left((y_2-\overline{m})-\beta^{\circ}_{2n}(y_1-\overline{m})\right)\right)'\left(\tau^{\circ}_{2n}\right)^{-1}\left((y_2-\overline{m})-\beta^{\circ}_{2n}(y_1-\overline{m})\right)\right) \\
&\leqslant 2\,(y_2-\overline{m})'\,\overline{\sigma}^{-1}_{\varpi}\,(y_2-\overline{m}) + 2\,\left(1+\lambda_{\mathsf{min}}(\varpi^{-1})\right)\left(1+\lambda_{\mathsf{min}}(\varpi)\right)^{-2}\,(y_1-\overline{m})'\,\overline{\sigma}^{-1}\,(y_1-\overline{m}).
\end{aligned} \tag{A.7}$$

The inequality (A.7) above together with (A.4) readily yields

$$\mathcal{S}^{\circ}_{2n}(y_1, dy_2) \geqslant \frac{1}{\sqrt{\det(2\pi\overline{\sigma})}}$$

$$\times \exp\left(-(1+\lambda_{\mathsf{min}}(\varpi^{-1}))(1+\lambda_{\mathsf{min}}(\varpi))^{-2}\ (y_1-\overline{m})'\,\overline{\sigma}^{-1}\,(y_1-\overline{m})\right)$$

$$\times\ \exp\left(-\,(y_2-\overline{m})'\,\overline{\sigma}_{\varpi}^{-1}\,(y_2-\overline{m})\right)\ dy_2.$$

Thus for any y_1 such that

$$\psi(y_1) = 1 + (y_1-\overline{m})'\overline{\sigma}^{-1}(y_1-\overline{m}) \leqslant r$$

we have

$$\mathcal{S}^{\circ}_{2n}(y_1, dy_2) \geqslant \varepsilon(r)\ \nu_{\overline{m},\overline{\sigma}_{\varpi}/2}(dy_2),$$

where

$$\varepsilon(r) = \frac{1}{\sqrt{\det(2\overline{\sigma}\overline{\sigma}_{\varpi}^{-1})}} \times \exp\left(-(1+\lambda_{\mathsf{min}}(\varpi^{-1}))(1+\lambda_{\mathsf{min}}(\varpi))^{-2}\ (r-1)\right).$$

This shows that the function ψ, and similarly the function φ defined in (5.10) satisfy the drift-minorization conditions stated in (4.6) and (4.7). ■

Lyapunov functions in (5.11) and (5.12). Using the triangle inequality for Mahalanobis distances we check that

$$\left[(\mathcal{Z}^{\circ}_{2n}(y)-\overline{m})'\overline{\sigma}^{-1}(\mathcal{Z}^{\circ}_{2n}(y)-\overline{m})\right]^{1/2}$$

$$\leqslant \left[(\beta^{\circ}_{2n}(y-\overline{m}))'\overline{\sigma}^{-1}(\beta^{\circ}_{2n}(y-\overline{m}))\right]^{1/2} + \left[((\tau^{\circ}_{2n})^{1/2}\ G)'\overline{\sigma}^{-1}((\tau^{\circ}_{2n})^{1/2}\ G)\right]^{1/2}$$

$$\leqslant (1+\lambda_{\mathsf{min}}(\varpi))^{-1}\ ((y-\overline{m})'\overline{\sigma}^{-1}(y-\overline{m}))^{1/2} + [G'G]^{1/2},$$

which yields the almost sure estimate

$$\left[(\mathcal{Z}^{\circ}_{2n}(y)-\overline{m})'\overline{\sigma}^{-1}(\mathcal{Z}^{\circ}_{2n}(y)-\overline{m})\right]^{1/2} \leqslant (1+\lambda_{\mathsf{min}}(\varpi))^{-1}\ ((y-\overline{m})'\overline{\sigma}^{-1}(y-\overline{m}))^{1/2} + \left[G'G\right]^{1/2}.$$

This implies that the function ψ, and similarly the function φ defined in (5.11) satisfy the drift-minorization conditions stated in (4.6) and (4.7).

The analysis of the Lyapunov functions (5.12) is based on the following technical lemma.

Lemma A.1. *For any matrices $\sigma > \tau$ and any $z \in \mathbb{R}^d$ and $0 \leqslant \delta \leqslant 1$ we have*

$$\mathbb{E}\left[\exp\left(\frac{\delta}{2}(z+\tau^{1/2}G)'\sigma^{-1}(z+\tau^{1/2}G)\right)\right] = \sqrt{\frac{\det(\sigma)}{\det(\sigma-\delta\tau)}}\ \exp\left(\frac{\delta}{2}z'(\sigma-\delta\tau)^{-1}z\right).$$

Proof. For any $g \in \mathbb{R}^d$ we have

$$\frac{\delta}{2}\ (z+g)'\sigma^{-1}(z+g) - \frac{1}{2}\ g'\tau^{-1}g = \frac{\delta}{2}\ z'\sigma^{-1}z + \frac{\delta^2}{2}\ z'\sigma^{-1}\chi\sigma^{-1}z - \frac{1}{2}\ (g-\delta\chi\sigma^{-1}z)'\chi^{-1}(g-\delta\chi\sigma^{-1}z),$$

with

$$0 < \chi := (\tau^{-1}-\delta\sigma^{-1})^{-1} = -\frac{1}{\delta}\ \sigma + \frac{1}{\delta}\sigma\tau^{-1}(\tau^{-1}-\delta\sigma^{-1})^{-1}.$$

We note that

$$\begin{aligned}\sigma^{-1}+\delta\sigma^{-1}\chi\sigma^{-1} &= \sigma^{-1}+\delta\sigma^{-1}\left(-\frac{1}{\delta}\ \sigma + \frac{1}{\delta}\sigma\tau^{-1}(\tau^{-1}-\delta\sigma^{-1})^{-1}\right)\sigma^{-1}\\ &= \tau^{-1}(\tau^{-1}-\delta\sigma^{-1})^{-1}\sigma^{-1} = (\sigma-\delta\tau)^{-1}\end{aligned}$$

and finally observe that

$$\frac{1}{\det(\tau\mathcal{X}^{-1})} = \frac{\det(\sigma)}{\det(\sigma-\delta\tau)}.$$

■

Using Lemma A.1 we check that

$$\mathbb{E}\left[\exp\left(\frac{\delta}{2}\left[(\mathcal{Z}^{\circ}_{2n}(y)-\overline{m})'\overline{\sigma}^{-1}(\mathcal{Z}^{\circ}_{2n}(y)-\overline{m})\right]\right)\right]$$

$$= \mathbb{E}\left[\exp\left(\frac{\delta}{2}\left[\left(\beta^{\circ}_{2n}(y-\overline{m})+(\tau^{\circ}_{2n})^{1/2}\ G\right)'\overline{\sigma}^{-1}\left(\beta^{\circ}_{2n}(y-\overline{m})+(\tau^{\circ}_{2n})^{1/2}\ G\right)\right]\right)\right]$$

$$= \frac{1}{\sqrt{\det(I-\delta\ \overline{\sigma}^{-1/2}\tau^{\circ}_{2n}\overline{\sigma}^{-1/2})}}\ \exp\left(\frac{\delta}{2}\ (y-\overline{m})'\ (\beta^{\circ}_{2n})'\,(\overline{\sigma}-\delta\ \tau^{\circ}_{2n})^{-1}\,\beta^{\circ}_{2n}\ (y-\overline{m})\right).$$

On the other hand, we have

$$\overline{\sigma}^{-1/2}\tau^{\circ}_{2n}\overline{\sigma}^{-1/2}\leqslant I$$

and arguing as above we conclude that

$$(\beta^{\circ}_{2n})'\,(\overline{\sigma}-\delta\ \tau^{\circ}_{2n})^{-1}\,\beta^{\circ}_{2n} = (\overline{\sigma}^{-1/2}\beta^{\circ}_{2n})'\left(I-\delta\ \overline{\sigma}^{-1/2}\tau^{\circ}_{2n}\overline{\sigma}^{-1/2}\right)^{-1}\overline{\sigma}^{-1/2}\beta^{\circ}_{2n} \tag{A.8}$$

$$\leqslant \frac{1}{1-\delta}\ (\beta^{\circ}_{2n})'\overline{\sigma}^{-1}\beta^{\circ}_{2n}\leqslant\frac{1}{1-\delta}(1+\lambda_{\mathsf{min}}(\varpi))^{-2}\ \overline{\sigma}^{-1}. \tag{A.9}$$

If we now consider the function

$$\psi(y) := \exp\left(\frac{\delta}{2}(y-\overline{m})'\overline{\sigma}^{-1}(y-\overline{m})\right)$$

then we have

$$\frac{1}{\psi(y)}\ \mathbb{E}\left(\psi\left(\mathcal{Z}^{\circ}_{2n}(y)\right)\right) \leqslant \frac{1}{\sqrt{\det(I-\delta\ I)}}\ \times$$
$$\times\ \exp\left(-\frac{\delta}{2}\ \left(1-\frac{1}{1-\delta}(1+\lambda_{\mathsf{min}}(\varpi))^{-2}\right)\ (y-\overline{m})'\ \overline{\sigma}^{-1}\ (y-\overline{m})\right).$$

Choosing

$$0<\delta<1-\frac{1}{(1+\lambda_{\mathsf{min}}(\varpi))^2}$$

we check that

$$\frac{1}{(1-\delta)(1+\lambda_{\mathsf{min}}(\varpi))^2}<1 \quad\text{and}\quad \lim_{|y|\to\infty}\frac{1}{\psi(y)}\ \mathcal{S}^{\circ}_{2n}(\psi)(y)=0.$$

This implies that the function ψ, and similarly the function φ, defined in (5.12) satisfy the drift-minorization conditions stated in (4.6) and (4.7).

Appendix B. Hilbert metric contractions

We recall the Hilbert contraction coefficient formula ([9, 12, 11, 66], see also [8, 25, 62])

$$\chi_H(\mathcal{K}) = \tanh\left(\frac{1}{4}\log\sup_{(x_1,x_2)\in\mathbb{X}^2} H(\delta_{x_1}\mathcal{K},\delta_{x_2}\mathcal{K})\right). \tag{B.1}$$